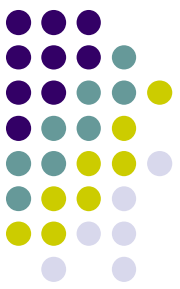


INTRODUCTION TO SYSTEMS ANALYSIS AND DESIGN: AN AGILE, ITERATIVE APPROACH

SATZINGER | JACKSON | BURD

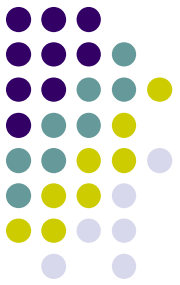
CHAPTER 2



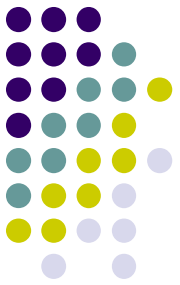
Chapter 2 Outline

- The RMO Consolidated Sales and Marketing System Project （RMO 综合销售项目）
- Systems Analysis Activities （系统分析活动）
- What Are Requirements? （什么是需求）
- Models and Modeling （模型和建模）
- Stakeholders （利益相关者）
- Information-Gathering Techniques （信息收集技术）
- Documenting Workflows with Activity Diagrams （用活动图记录工作流）

Learning Objectives

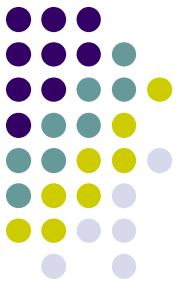


- Describe the activities of systems analysis (描述系统分析的活动)
- Explain the difference between **functional** and **nonfunctional** requirements (解释功能性需求和非功能性需求之间的区别)
- Describe the role of models in systems analysis (描述模型在系统分析中的作用)



Learning Objectives

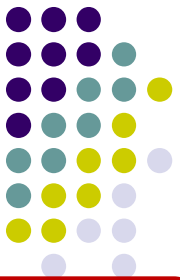
- Identify and understand different kinds of stakeholders and their contributions to requirements definition (利益相关者对需求的不同贡献)
- Describe information-gathering techniques and determine when each is best applied (信息收集技术)
- Develop activity diagrams to model workflows (开发活动图，建模工作流)



Chapter 2 Outline

- The RMO Consolidated Sales and Marketing System Project (RMO 综合销售和市场营销系统项目)
- Systems Analysis Activities
- What Are Requirements?
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- Stakeholders
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RMO Existing Application Architecture (RMO 现有结构)



- Supply Chain Management (SCM)
 - 5 years old; Java/Oracle
 - Tradeshow system will interface with SCM

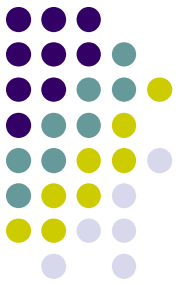
问题：供应链的管理方式发生变化
- Phone/Mail Order System
 - 12 years old; Visual Studio/SQL
 - Reached capacity;

问题：购物方式发生变化
- Retail Store System
 - Older package solution;

问题：购物模式发生变化
- Customer Support System (CSS)
 - Web based system; evolved over the years, mainly support phone/mail orders

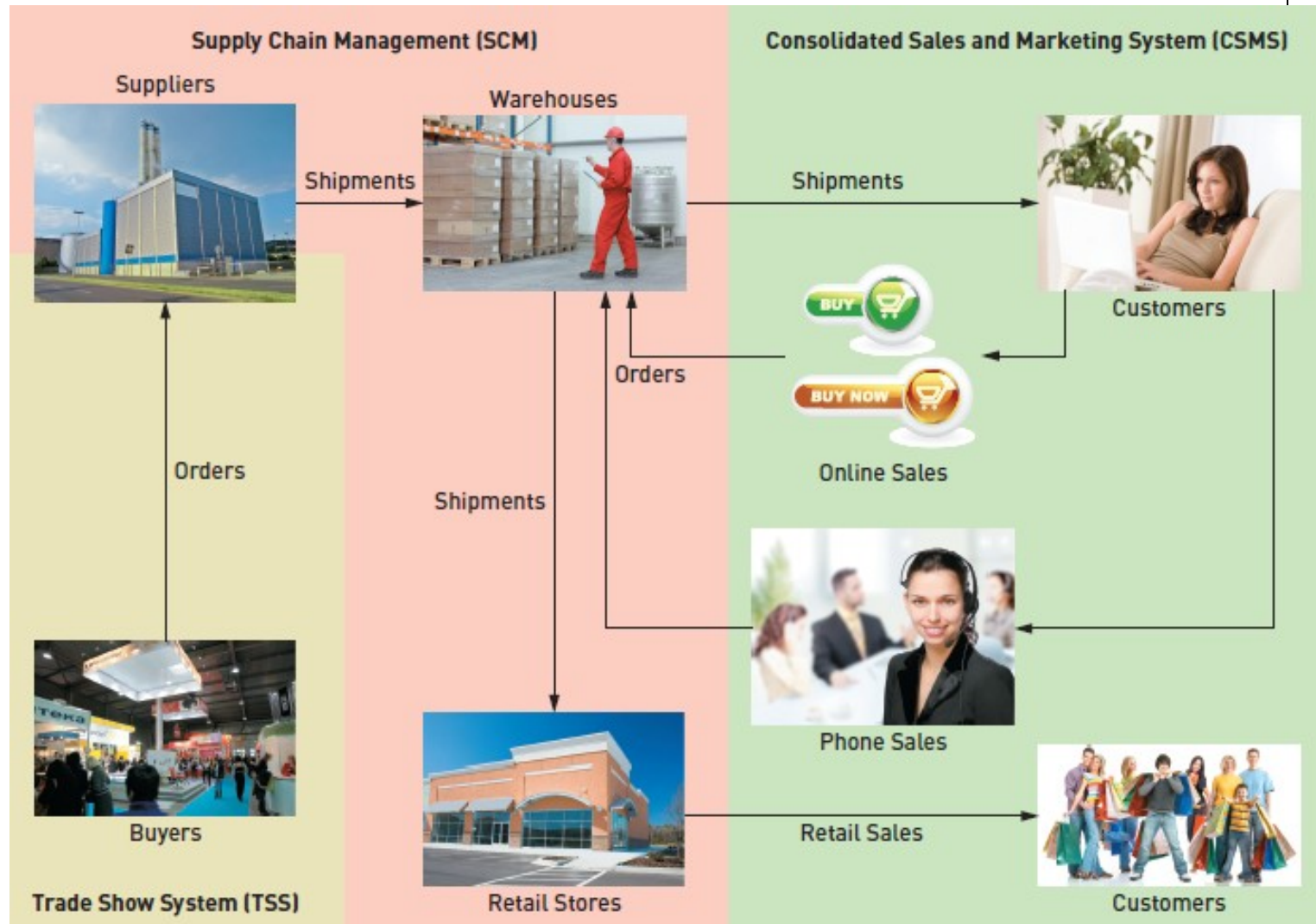
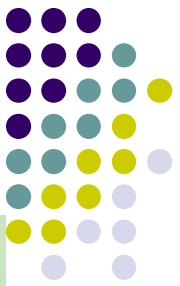
问题：顾客需求发生变化

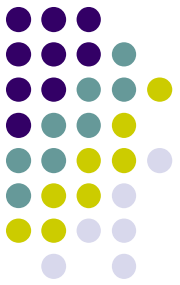
New Consolidated Sales and Marketing System (CSMS , 新的市场推广系统)



- Sales Subsystem (销售子系统)
 - Integrates online, phone, and retail stores
- Order Fulfillment Subsystem (订单实施子系统)
 - Track shipments, rate products and services
- Customer Account Subsystem (顾客账务子系统)
 - Shopping history, linkups, “mountain bucks” rewards
- Marketing Subsystem (市场营销子系统)
 - Promotional packages, partner relationships, more complete merchandise information and reporting

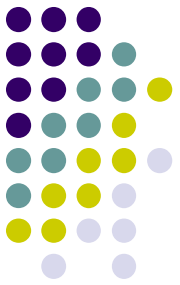
Proposed **Application Architecture:** Integrate SCM and New CSMS





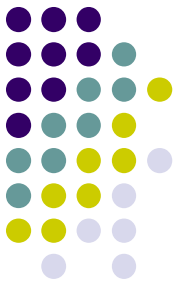
Chapter 2 Outline

- The RMO Consolidated Sales and Marketing System Project
- **Systems Analysis Activities** (系统分析活动)
- What Are Requirements?
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- Information-Gathering Techniques
- Documenting Workflows with Activity Diagrams



Systems Analysis Activities

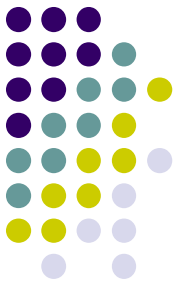
- Gather Detailed Information (收集信息)
 - Interviews, questionnaires, documents, observing business processes, researching vendors, comments and suggestions
- Define Requirements (定义需求)
 - Modeling functional requirements and non-functional requirements
- Prioritize Requirements (优先需求)
 - Essential, important, vs. nice to have
- Develop User-Interface Dialogs (用户界面对话框)
 - Flow of interaction between user and system
- Evaluate Requirements with Users (与用户评估需求)
 - User involvement, feedback, adapt to changes



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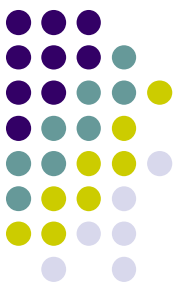
What Are Requirements?



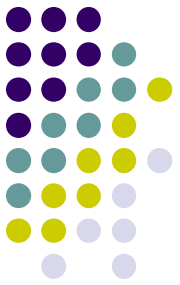
- System Requirements (系统需求) =
 - Functional requirements (功能需求)
 - Non-functional requirements (非功能需求)
- Functional Requirements (功能需求) — the activities the system **must** perform (系统必须执行的活动)
 - Business uses, functions the users carry out
 - Use cases in Chapter 1
- Non-Functional Requirements— other system characteristics (非功能需求——其他系统特征)
 - Constraints and performance goals

FURPS+ Requirements Acronym

识别和分类需求的方法—— FURPS+

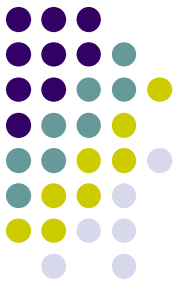


- Functional requirements (功能需求)
- Usability requirements (可用性需求)
- Reliability requirements (可靠性需求)
- Performance requirements (性能需求)
- Security requirements (安全需求)
- + even more categories...



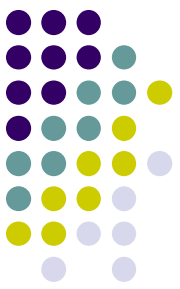
FURPS+ Requirements Acronym

Requirement categories	FURPS + categories	Example requirements
Functional	Functions	Business rules and processes
Nonfunctional 可用性 可靠性 性能 安全 设计限制 开发 接口 物理层 支持	Usability Reliability Performance Security + Design constraints Implementation Interface Physical Support	User interface, ease of use Failure rate, recovery methods Response time, throughput Access controls, encryption Hardware and support software Development tools, protocols Data interchange formats Size, weight, power consumption Installation and updates



Chapter 2 Outline

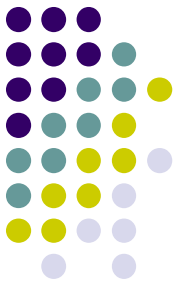
- The RMO Consolidated Sales and Marketing System Project
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- **Models and Modeling (模型和建模)**
- Stakeholders
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Models and Modeling

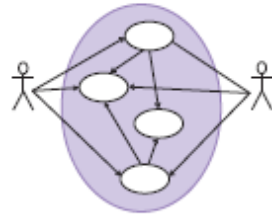
- How to **represent** the requirements? 如何表示需求
- Model— a representation of some aspect of the system being built
- Types of Models (模型类型)
 - Textual model (文本) — something written down, described
 - Graphical models (图形) — diagram, schematic
 - Mathematical models (数学公式) — formulas, statistics, algorithms
- Unified Modeling Language (UML , 统一建模语言)
 - Standard graphical modeling symbols/terminology used for information systems

Some Analysis and Design Models



1 buy new car
2 sell car
3 get car serviced
4 make payment
5 trade in car

Event list



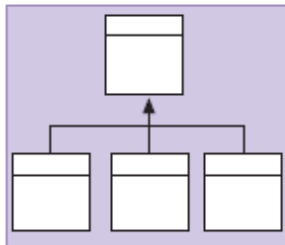
Use case diagram



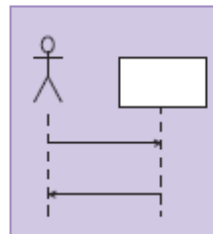
Use case description



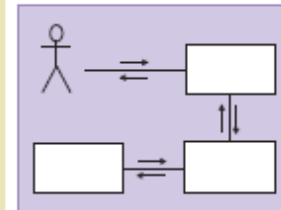
Location diagram



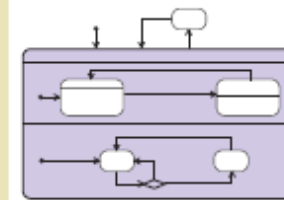
Class diagram



Sequence diagram

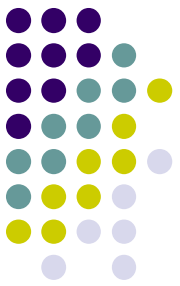


Communication diagram

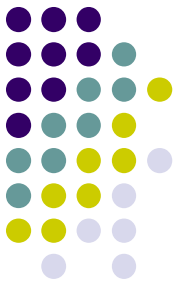


State machine diagram

Reasons for Modeling (建模的原因)



- Learning from the modeling process (从建模过程中学习)
- Reducing complexity by abstraction (通过抽象降低复杂性)
- Remembering all the details (记住所有的细节)
- Communicating with other development team members (与其他成员 / 相关人员沟通)
- Documenting what was done for future maintenance/enhancement (维护、记录)

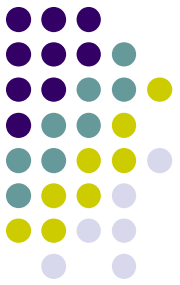


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- Documenting Workflows with Activity Diagrams

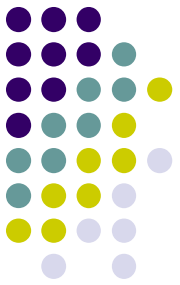
Stakeholders

Who do you involve and talk to?



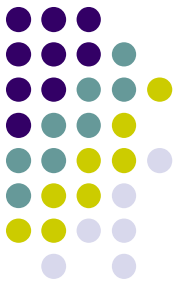
- **Stakeholders** (利益相关者) — persons who have an interest in the successful implementation of the system
- **Internal Stakeholders** (内部利益相关者) — persons within the organization
- **External stakeholders** (外部利益相关者) — persons outside the organization
- **Operational stakeholders** (操作相关者) — persons who regularly interact with the system
- **Executive stakeholders** (执行相关者) — persons who don't directly interact, but use the information or have financial interest

Stakeholders of a comprehensive accounting system for public company



Stakeholders

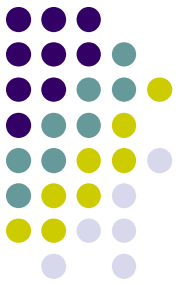
For RMO CSMS Project



- Phone/mail sales order clerks （电话 / 邮件订单员）
- Warehouse and shipping personnel （仓库运输人员）
- Marketing personnel who maintain online catalog information （维护在线目录的营销人员）
- Marketing, sales, and financial managers （市场营销）
- Senior executives （高级管理人员）
- Customers （客户）
- External shippers (e.g., UPS and FedEx) （外部托运人）

Stakeholders

For RMO CSMS Project



Operational

Executive

- Order clerks
- Warehouse and shipping personnel
- Marketing personnel who maintain online catalog information
- Marketing, sales, and financial managers
- Senior executives
- Customers
- External shippers (e.g., UPS and FedEx)

Internal

- Marketing personnel who maintain online catalog information
- Marketing, sales, and financial managers

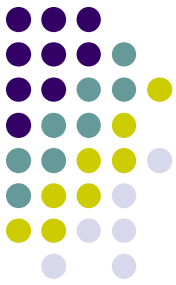
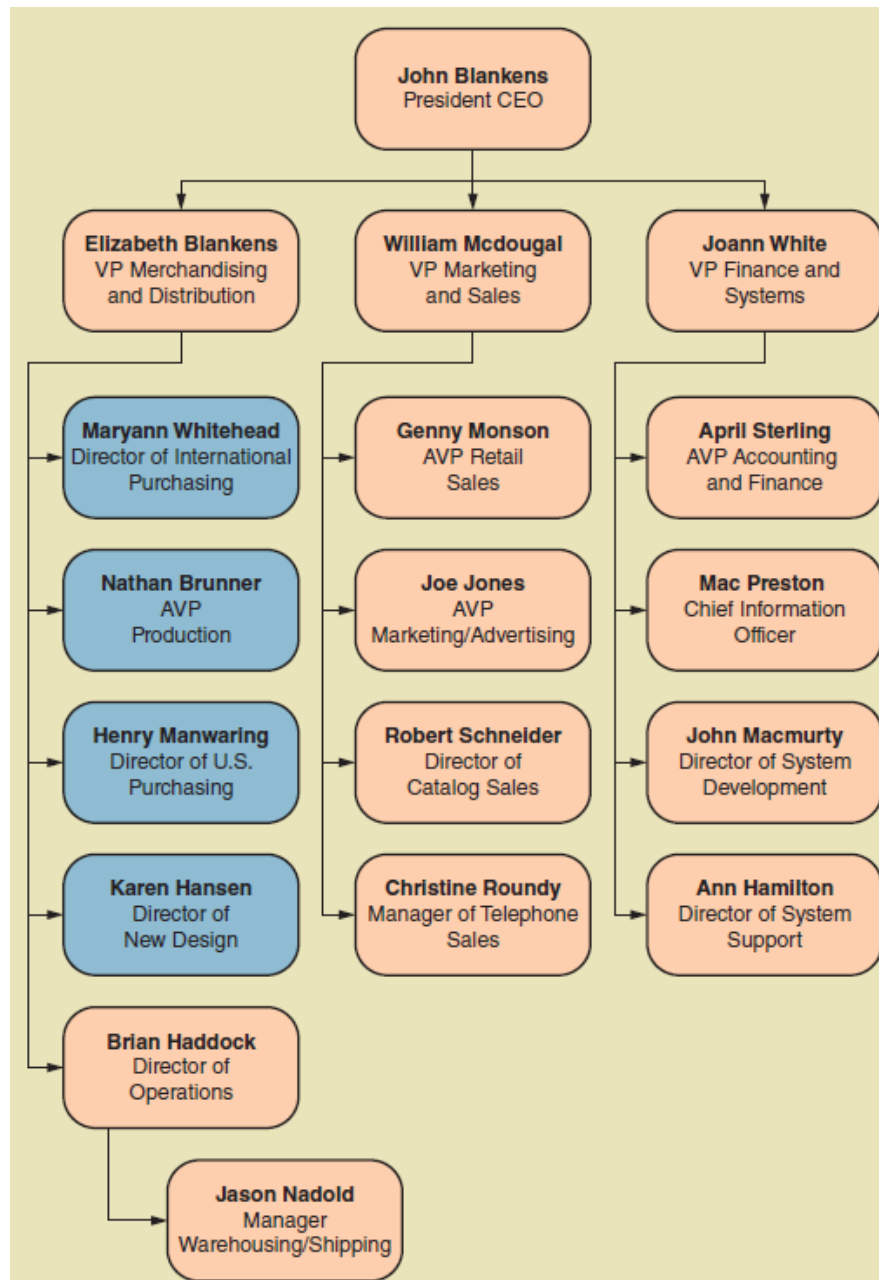
- Senior executives
- Warehouse and shipping personnel

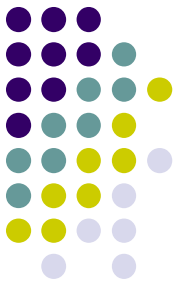
External

- Order clerks
- Customers

- External shippers

RMO Internal Stakeholders

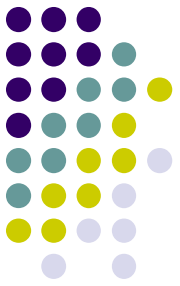




Chapter 2 Outline

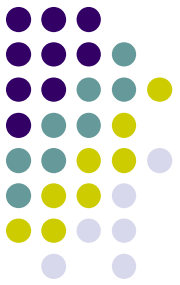
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Information Gathering Techniques



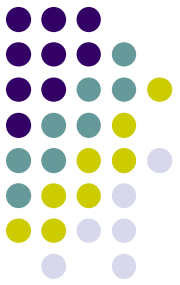
- Interviewing users and other stakeholders (采访用户和其他利益相关者)
- Distributing and collecting questionnaires (发放问卷)
- Reviewing inputs, outputs, and documentation (审查输入、输出和文档)
- Observing and documenting business procedures (观察和记录业务流程)
- Researching vendor solutions (研究供应商解决方案)
- Collecting active user comments and suggestions (收集活跃用户的意见和建议)

Interviewing Users and Other Stakeholders (采访用户)



- Prepare detailed questions (准备详细的问题)
- Obtain and discuss answers to the questions (讨论并获取答案)
- Document the answers (整理答案)
- Follow up as needed in future meetings or interviews (后续跟进)

Themes for Information Gathering Questions



Theme	Questions to users
What are the business operations and processes?	What do you do?
How should those operations be performed?	How do you do it? What steps do you follow? How could they be done differently?
What information is needed to perform those operations?	What information do you use? What inputs do you use? What outputs do you produce?

Distribute and Collect Questionnaires

RMO Questionnaire

This questionnaire is being sent to all telephone-order sales personnel. As you know, RMO is developing a new customer support system for order taking and customer service.

The purpose of this questionnaire is to obtain preliminary information to assist in defining the requirements for the new system. Follow-up discussions will be held to permit everybody to elaborate on the system requirements.

Part I. Answer these questions based on a typical four-hour shift.

1. How many phone calls do you receive? _____
2. How many phone calls are necessary to place an order for a product? _____
3. How many phone calls are for information about RMO products, that is, questions only? _____
4. Estimate how many times during a shift customers request items that are out of stock. _____
5. Of those out-of-stock requests, what percentage of the time does the customer desire to put the item on back order? _____%
6. How many times does a customer try to order from an expired catalog? _____
7. How many times does a customer cancel an order in the middle of the conversation? _____
8. How many times does an order get denied due to bad credit? _____

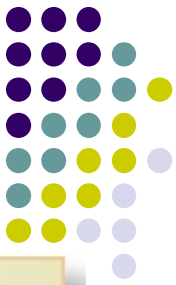
Part II. Circle the appropriate number on the scale from 1 to 7 based on how strongly you agree or disagree with the statement.

Question	Strongly Agree				Strongly Disagree			
It would help me do my job better to have longer descriptions of products available while talking to a customer.	1	2	3	4	5	6	7	
It would help me do my job better if I had the past purchase history of the customer available.	1	2	3	4	5	6	7	
I could provide better service to the customer if I had information about accessories that were appropriate for the items ordered.	1	2	3	4	5	6	7	
The computer response time is slow and causes difficulties in responding to customer requests.	1	2	3	4	5	6	7	

Part III. Please enter your opinions and comments.

Please briefly identify the problems with the current system that you would like to see resolved in a new system.

Review Inputs, Outputs, and Procedures





Ridgeline Mountain Outfitters—Customer Order Form

Name and address of person placing order.
(Please verify your mailing address and make correction below.)
Order Date ____/____/____

Name _____
Address _____ Apt. No. _____
City _____ State _____ Zip _____
Phone: Day () _____ Evening () _____

Gift Order or Ship To: (Use only if different from address at left.)
Name _____
Address _____ Apt. No. _____
City _____ State _____ Zip _____
Gift ☐ Address for this Shipment Only ☐ Permanent Change of Address ☐
Gift Card Message _____
Delivery Phone () _____

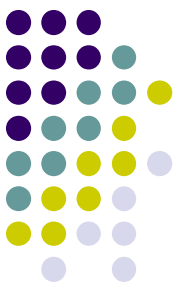
Item No.	Description	Style	Color	Size	Sleeve Length	Qty	Monogram	Style	Price Each	Total

Method of Payment

Check/Money Order ☐ Gift Certificate(s) ☐ AMOUNT ENCLOSED \$ _____
American Express ☐ MasterCard ☐ VISA ☐ Other ☐
Account Number _____ MO YR _____
____/____
Expiration Date

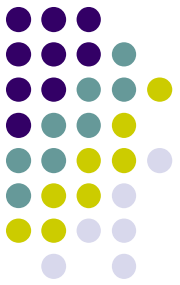
Signature _____

MERCHANDISE TOTAL _____
Regular FedEx shipping \$4.50 per U.S. delivery address
(Items are sent within 24 hours for delivery in 2 to 4 days)
Please add \$4.50 per each additional U.S. delivery address _____
FedEx Standard Overnight Service _____
Any additional freight charges _____
International Shipping (see shipping information on back) _____



Additional Techniques

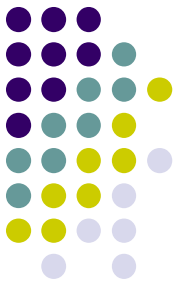
- Observe and Document Business Processes (观察和记录业务流程)
 - Watch and learn
 - Document with Activity diagram (next section)
- Research Vendor Solutions (研究供应商解决方案)
 - See what others have done for similar situations
 - White papers, vendor literature, competitors
- Collect Active User Comments and Suggestions (收集用户反馈)
 - Feedback on models and tests



Chapter 2 Outline

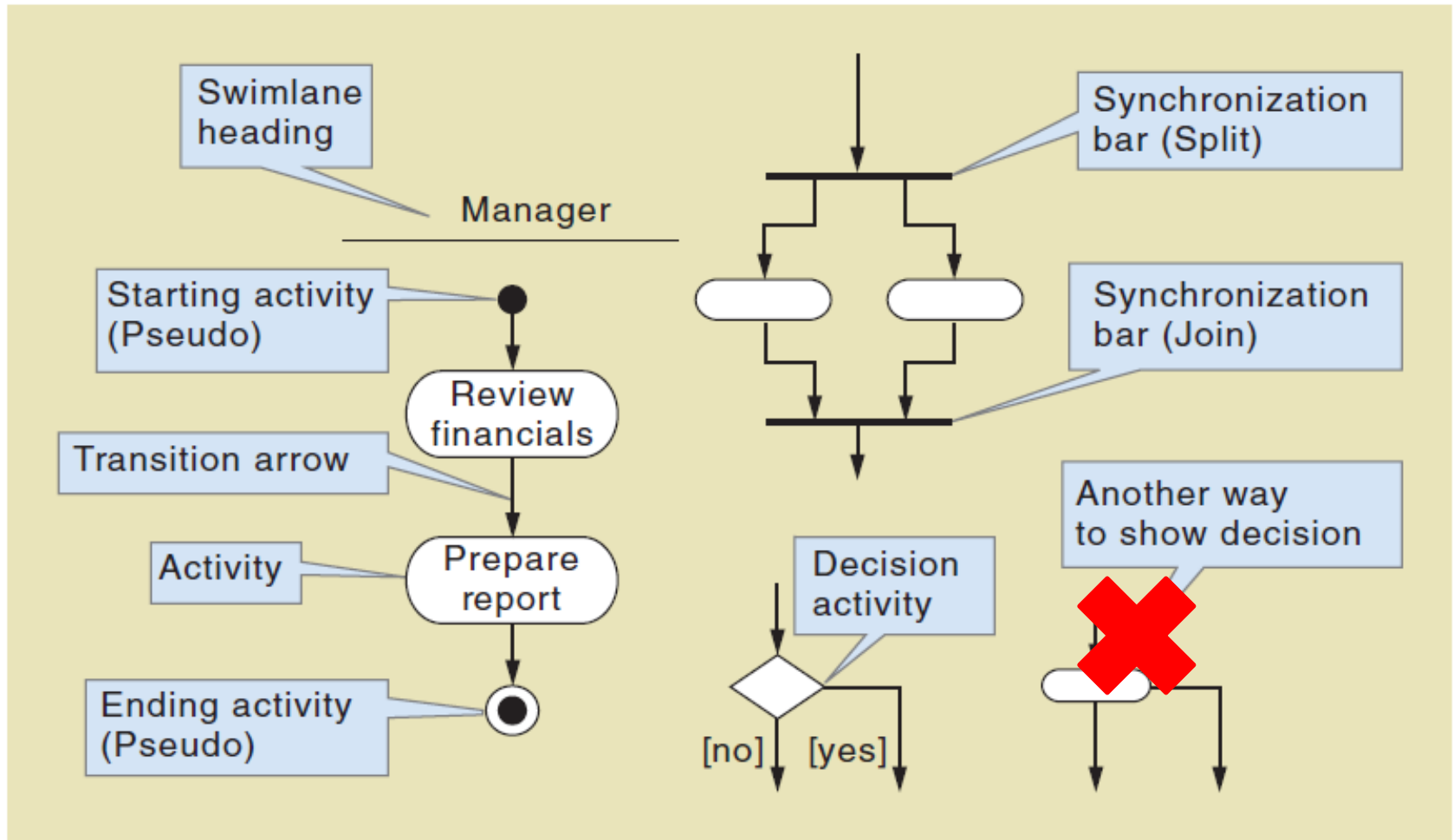
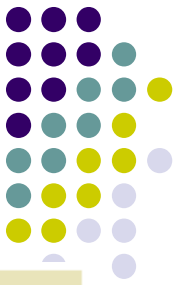
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Documenting Workflows with Activity Diagrams

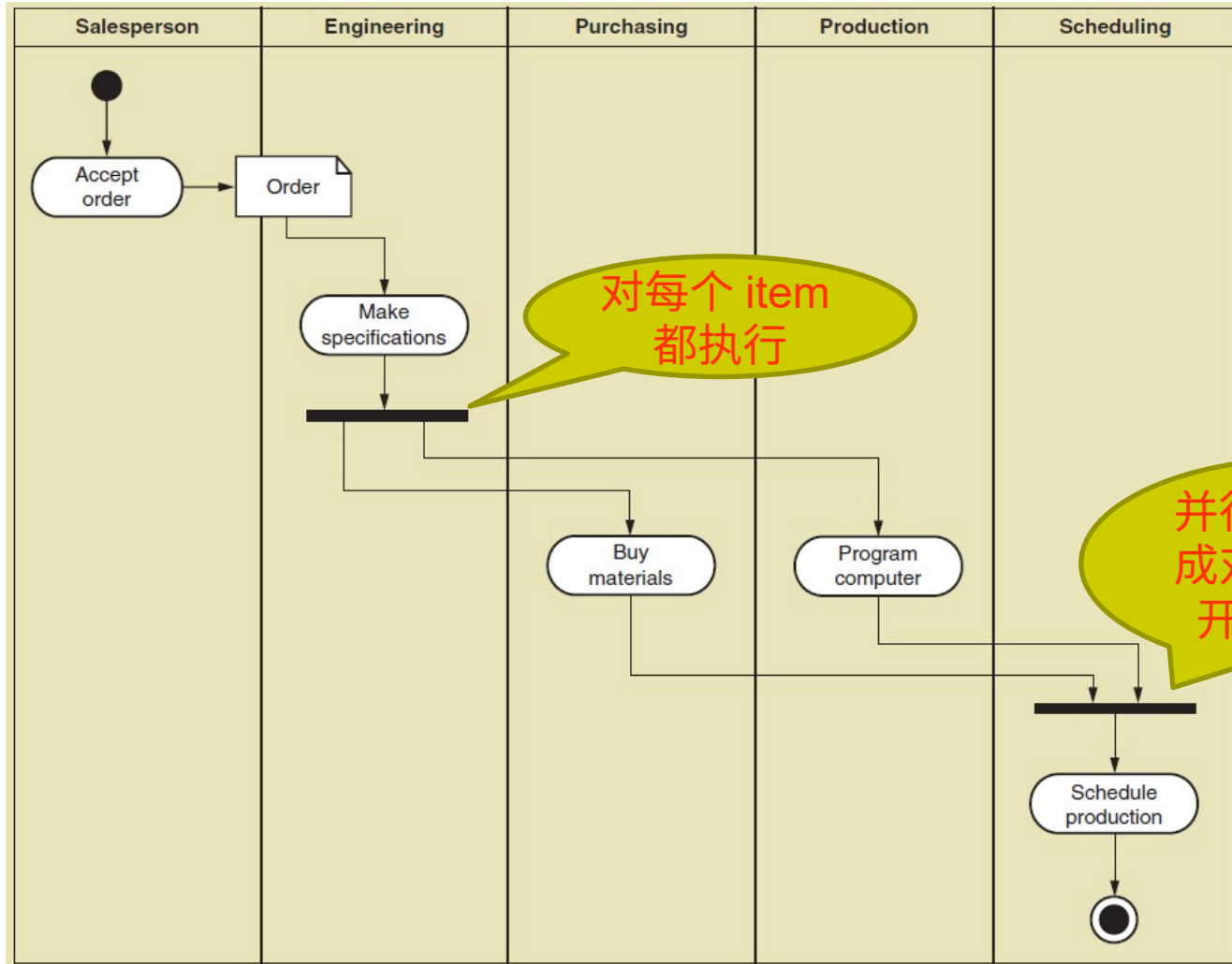
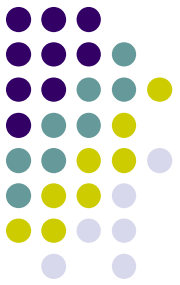


- **Workflow (工作流)** — sequence of processing steps that completely handles one business transaction or customer request (完成处理一个业务的一系列处理步骤)
- **Activity Diagram (活动图)** — describes user (or system) activities, the person who does each activity, and the sequential flow of these activities
 - Useful for showing a graphical model of a workflow (用于显示工作流的图形化模型)
 - A UML diagram

Activity Diagrams Symbols



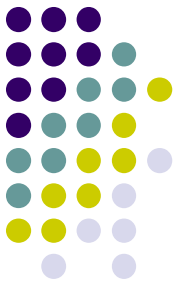
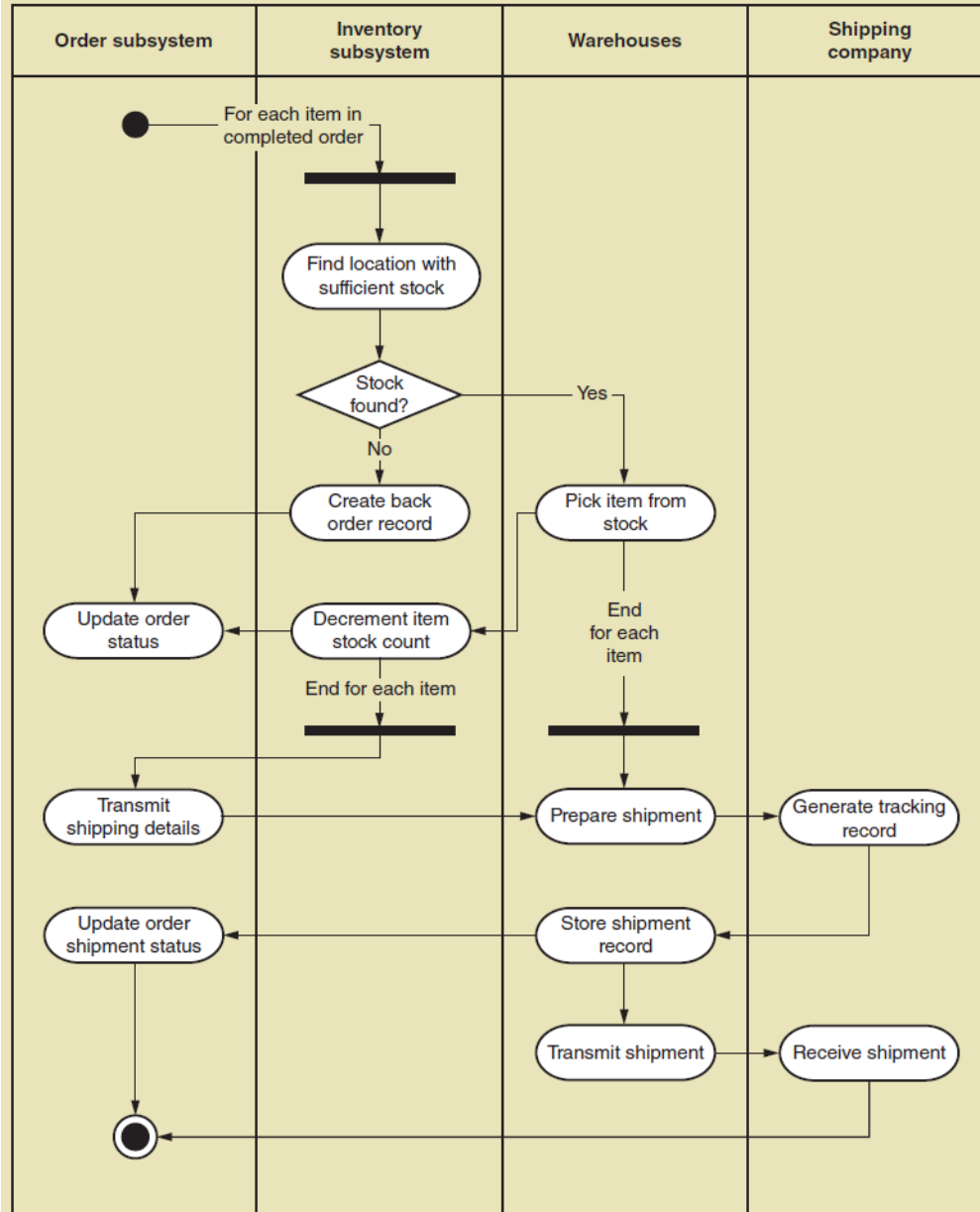
Activity Diagram with Concurrent Paths



Introduction to Systems Analysis and Design, 6th Edition

Activity Diagram for RMO Order Fulfillment

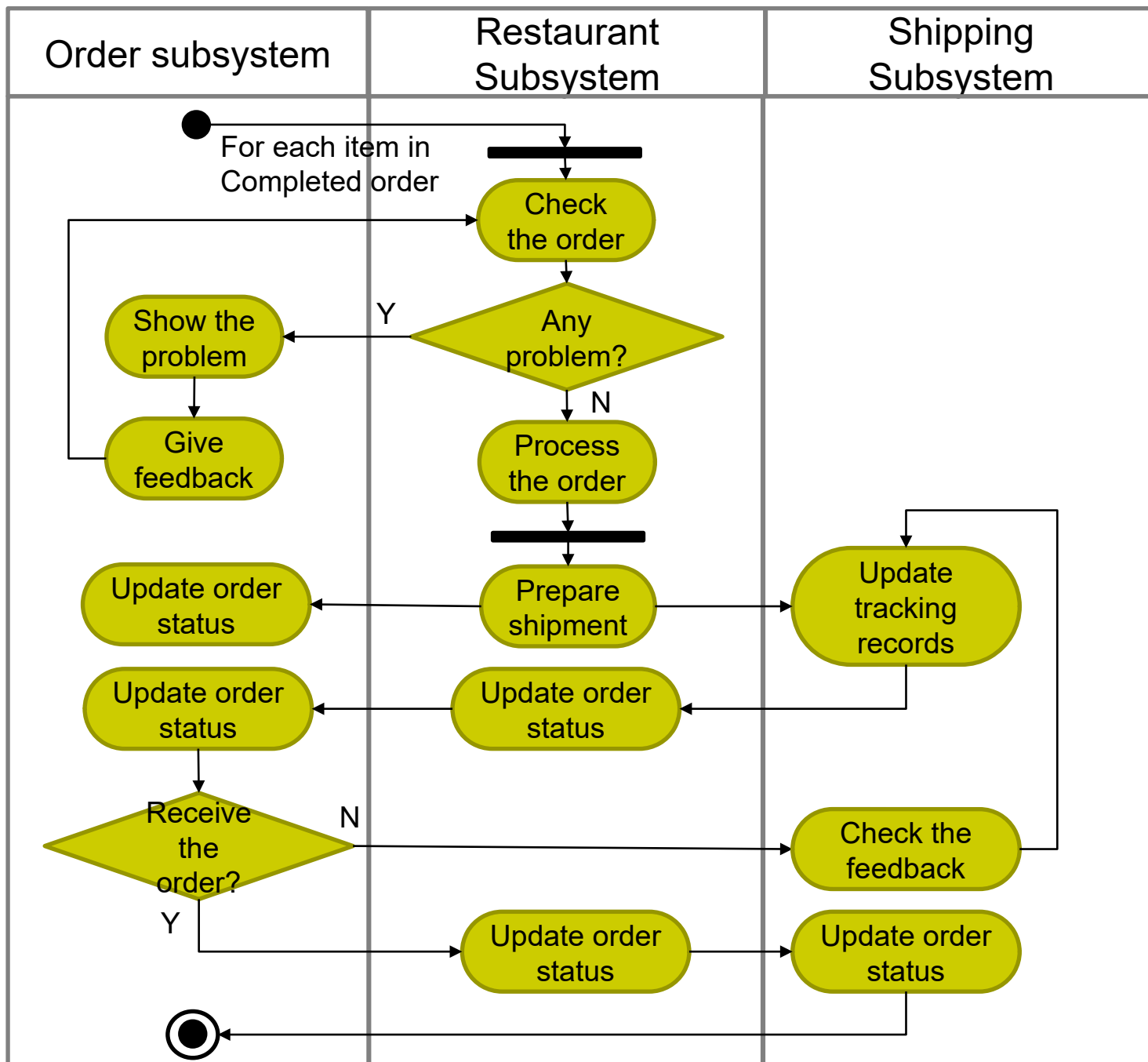
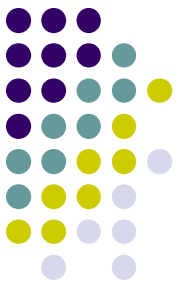
Order Fulfillment

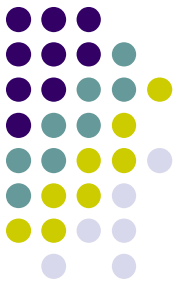




Please analyze the process of ordering a delivery(外卖) from a restaurant website, and then draw an Activity Diagram in UML to reasonably show this process.

正常使用主观题需 2.0 以上版本雨课堂





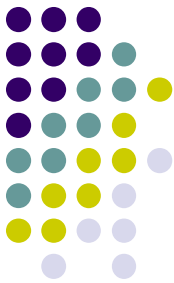
补充内容：结构化需求分析方法

1. 结构化分析方法概述

1) 分析策略

结构化分析（Structured Analysis，简称 SA）方法是 20 世纪 70 年代由 E·Yourdon 等人提出的一种面向数据流的分析方法，适用于大型的数据处理系统。由于利用图形来表达需求会使文档清晰、简明、易于学习和掌握，所以软件分析人员仍在广泛使用这种传统的分析方法。

结构化分析方法总的指导思想是“自顶向下，逐步求精”，它的两个基本原则是“抽象”和“分解”，即按照功能分解的原则，对系统进行逐层分解，直到找到所有满足功能要求的可实现软件元素为止。



补充内容：结构化需求分析方法

1. 结构化分析方法概述

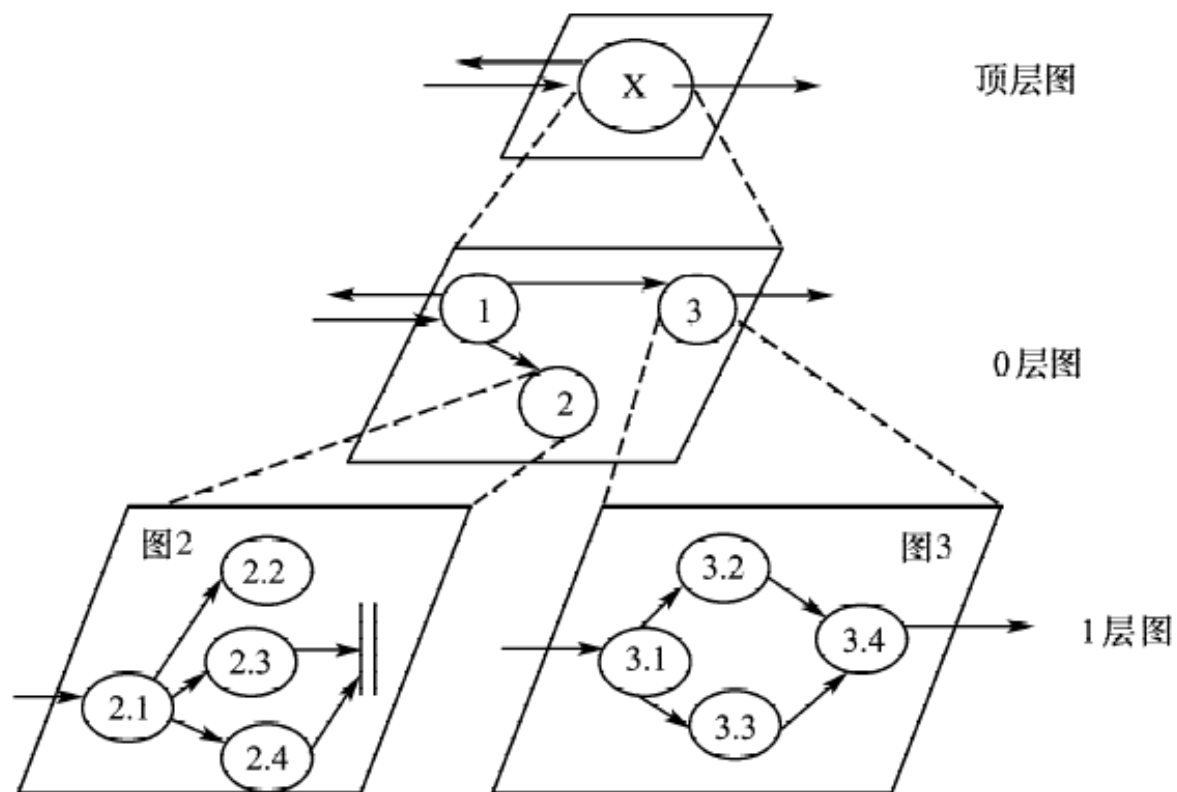
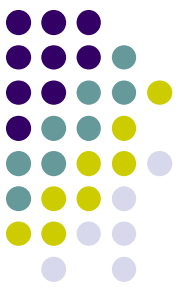


图 3-2 系统分层数据流图



补充内容：结构化需求分析方法

1. 结构化分析方法概述

2) 描述工具

结构化分析方法利用图形等半形式化的描述表达需求，用它们形成需求规格说明书的主要部分，主要工具有：

（1）数据流图（DFD）。描述系统的分解，即描述系统由哪几部分组成，各部分之间有什么联系等。

（2）数据词典（DD）。明确定义数据流图中的数据和加工。它是数据流条目、数据存储条目、数据项条目和基本加工条目的汇集。

（3）结构化语言、判定表和判定树。用于详细描述数据流图中不能再分解的每一个基本加工的处理逻辑。



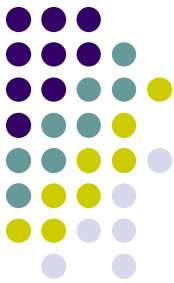
补充内容：结构化需求分析方法

(1) 数据流图

1. 数据流图的基本符号

表 3-1 数据流图中的基本图形符号

符号	说明
	圆或椭圆。表示加工,也称数据处理,它对数据流进行某些操作或变换。应在圆圈内写上加工名字(一般为动词短语),在分层数据流图中,加工还应编号
	矩形。表示数据流的源点和终点,是软件系统外部环境中的实体(包括人员、组织或其他软件系统),统称外部实体。应在方框内写上相应的名称
	双杠。表示文件或称数据存储,暂时保存数据,它可以是数据库文件或任何形式的数据组织。指向存储的数据可理解为写入,从存储引出的数据可理解为读出。要用名词或名词短语进行命名
	箭头。表示数据流,由一组成分固定的数据项组成。数据流必须有流向,箭头表示数据流动的方向,即在加工之间、加工与源点/终点之间、加工与数据存储之间流动。除与数据存储之间的数据流不用命名外,其他数据流均需用名词或名词短语命名



补充内容：结构化需求分析方法

（1）数据流图

2. 数据流图的绘制步骤

（1）画顶层数据流图

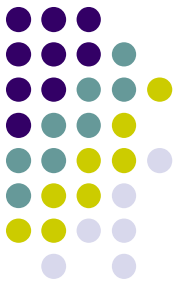
列出系统的全部数据源点和终点，将系统加工处理过程作为一个整体，就可能得到顶层图。具体说就是：画一个圆，在其中写上系统名称，然后在圆的外围画上系统的输入和输出，这一步工作实际上是决定研究的内容和系统的范围。

（2）画各层数据流图

对系统处理过程自顶向下，逐步分解，画出各层的数据流图。

（3）画总的数据流图

这一步对了解整个系统很有好处，但也要根据实际情况来决定总图的布局，不要把数据流图画得太复杂。



补充内容：结构化需求分析方法

（1）数据流图

3. 数据流图中的命名规则

（1）数据流

数据流表明数据和数据流向，它通常由一组数据项组成。

（2）加工

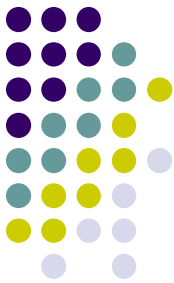
加工是对数据的某种操作或变换。

（3）文件

文件起暂时保存数据的作用。

（4）数据源点和终点

数据源点和终点是数据的始发点和终止点，是软件系统外部环境中的实体（包括人员、组织或其他软件系统），统称外部实体。



补充内容：结构化需求分析方法

（1）数据流图

4. 数据流图中分层技术

对于比较复杂的实际问题，在数据流图上常常出现十几个乃至几十个、上百个加工，这样的数据流图复杂而且难以理解。为了避免这种情况出现，可以采用数据流图的分层技术。分层技术的基本思想是，不是在一个数据流图中一次引入太多的细节，而是有控制地逐步增加细节，实现从抽象到具体的逐步过渡。

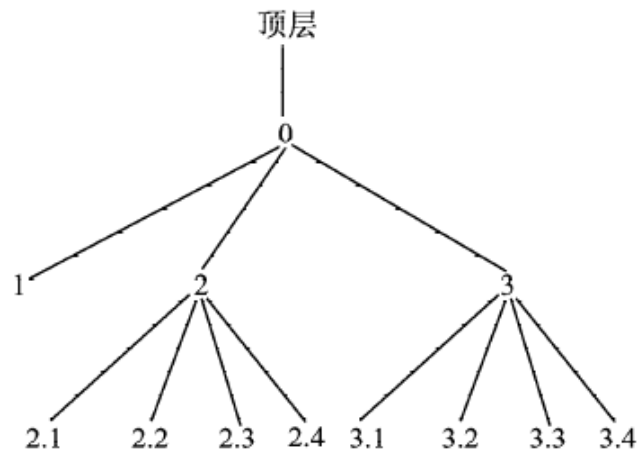
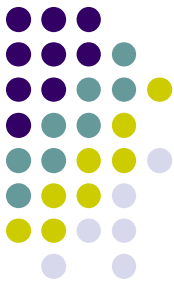


图 3-6 各数据流图之间的关系

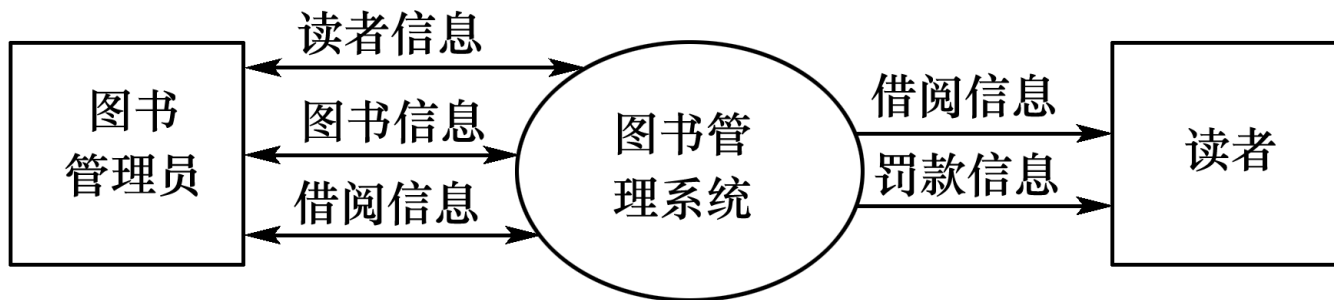


补充内容：结构化需求分析方法

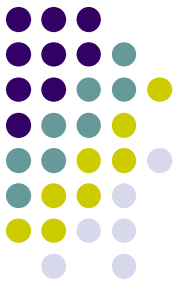
（1）数据流图

下面以第 2 章的“图书管理系统”为例，绘制数据流图。

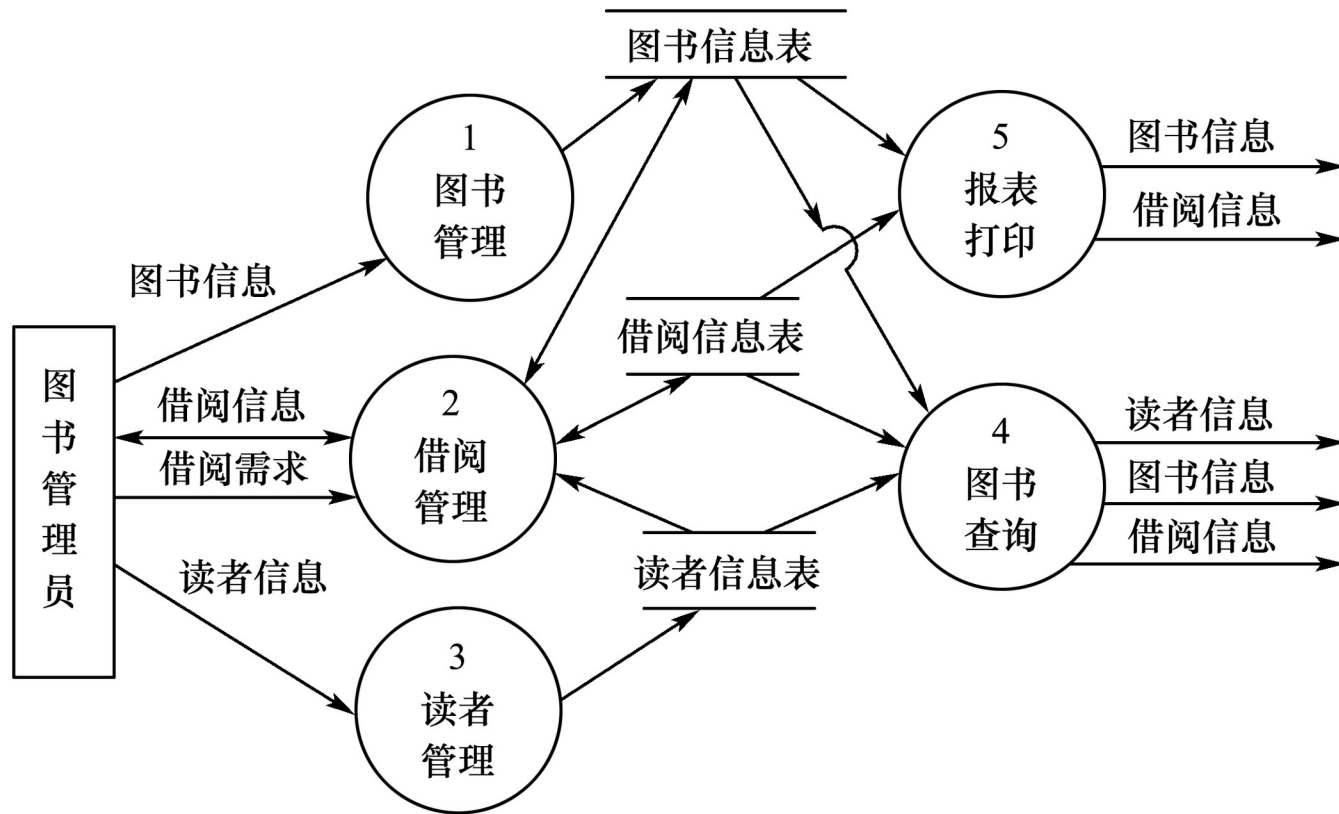
在图书馆负责人和计算机系的技术人员通过了“图书管理系统”《项目开发计划》后，项目组随即进入了项目开发阶段，计算机系教师与图书馆相关业务人员紧密合作，经过 15 天的工作，形成了“图书管理系统”《软件需求说明书》。主要数据流图如下：



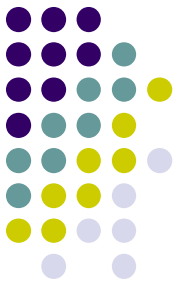
(a)顶层数据流图



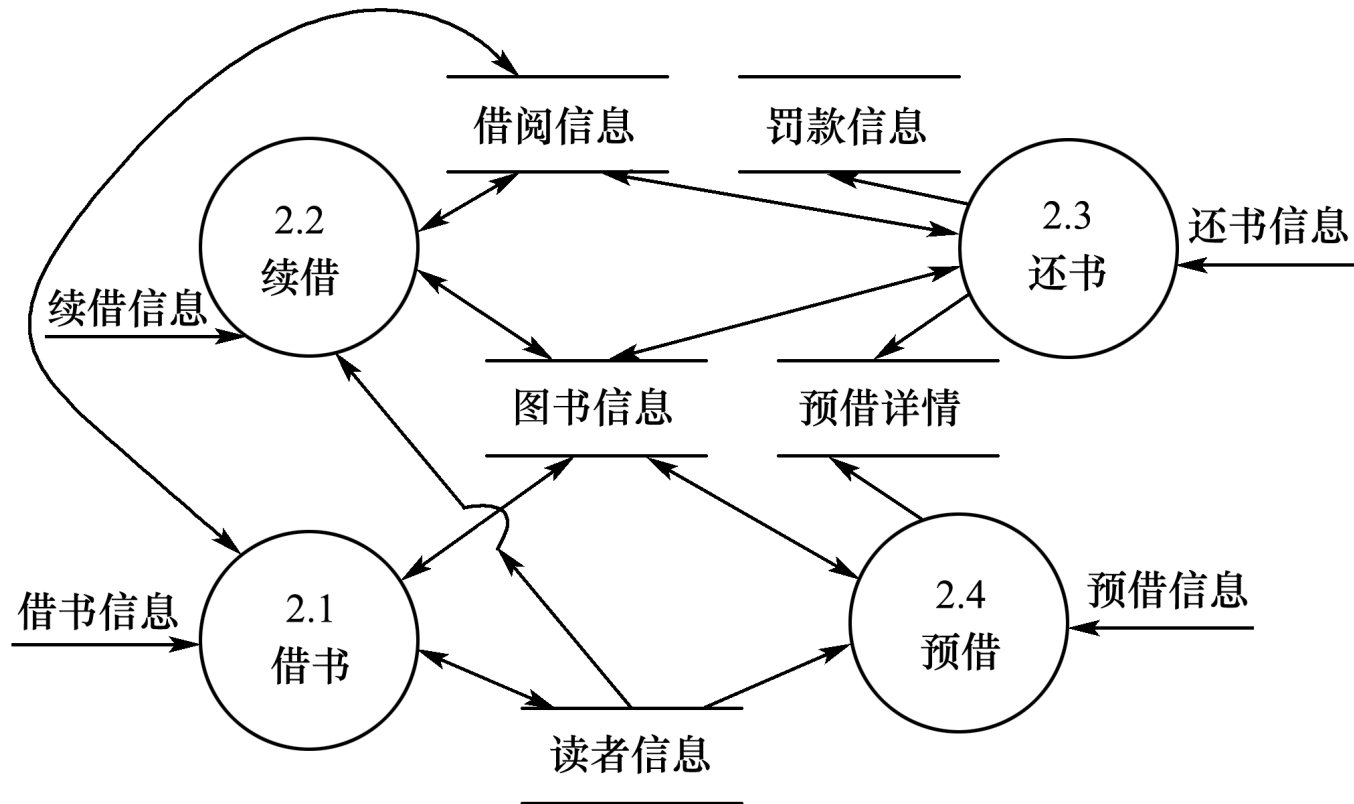
补充内容：结构化需求分析方法



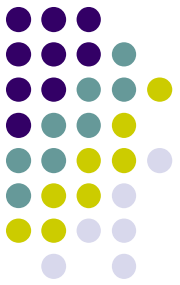
(b)0层数据流图



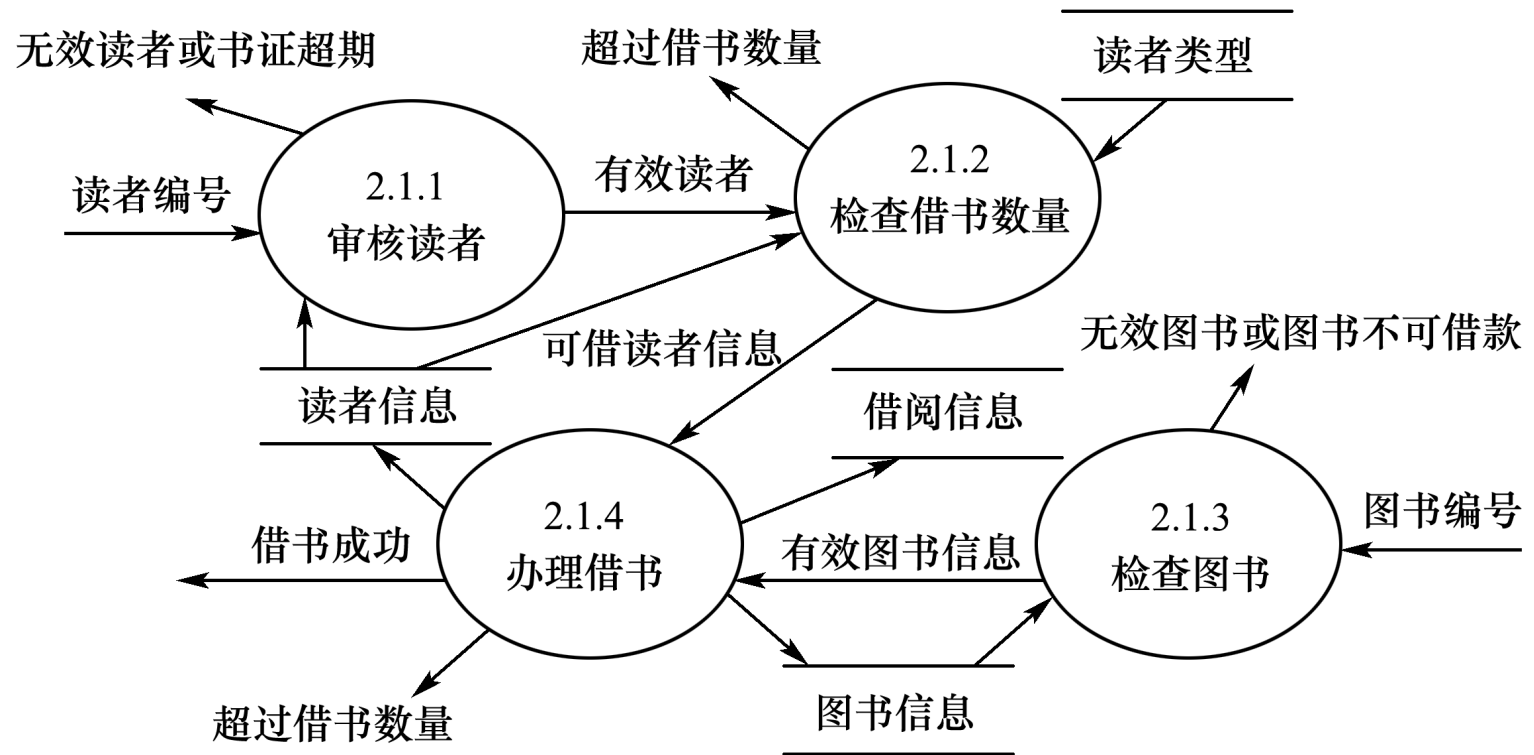
补充内容：结构化需求分析方法



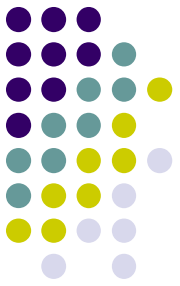
(c) 借阅管理加工的一层数据流图



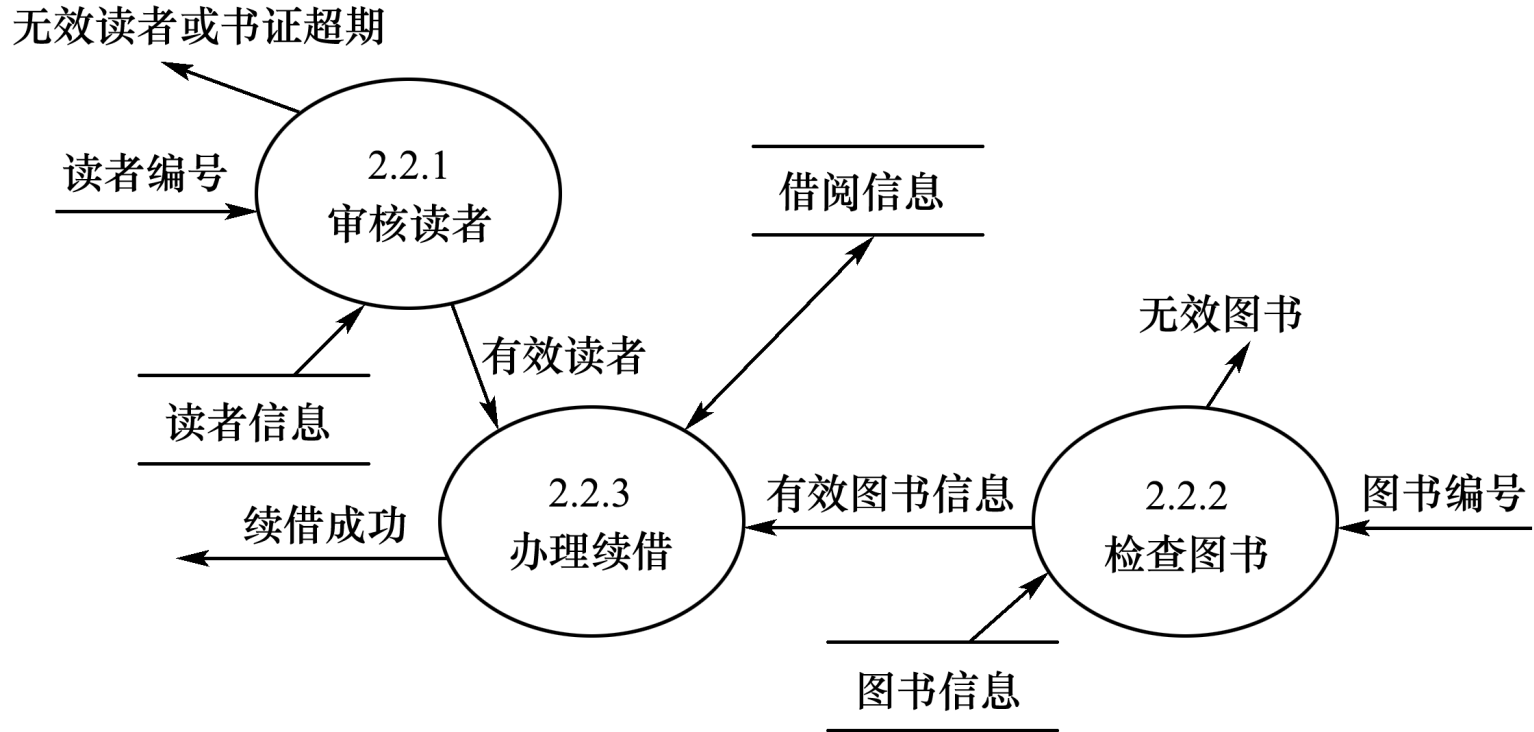
补充内容：结构化需求分析方法



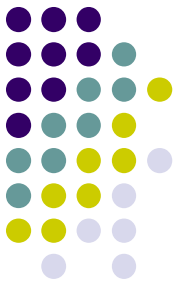
(d) 借阅管理加工的二层数据流图(借书)



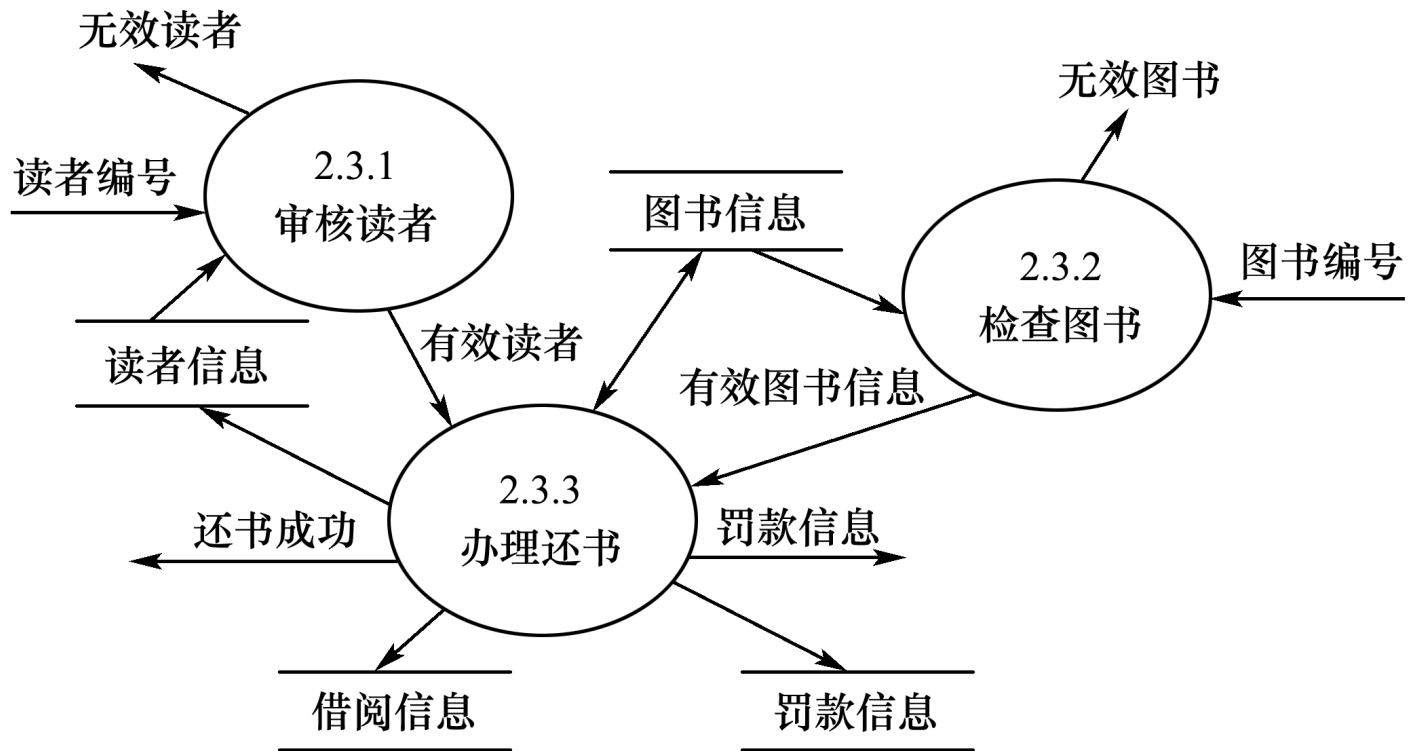
补充内容：结构化需求分析方法



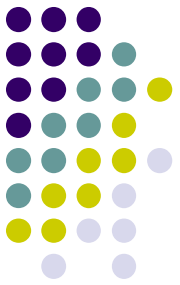
(e) 借阅管理加工的二层数据流图(续借)



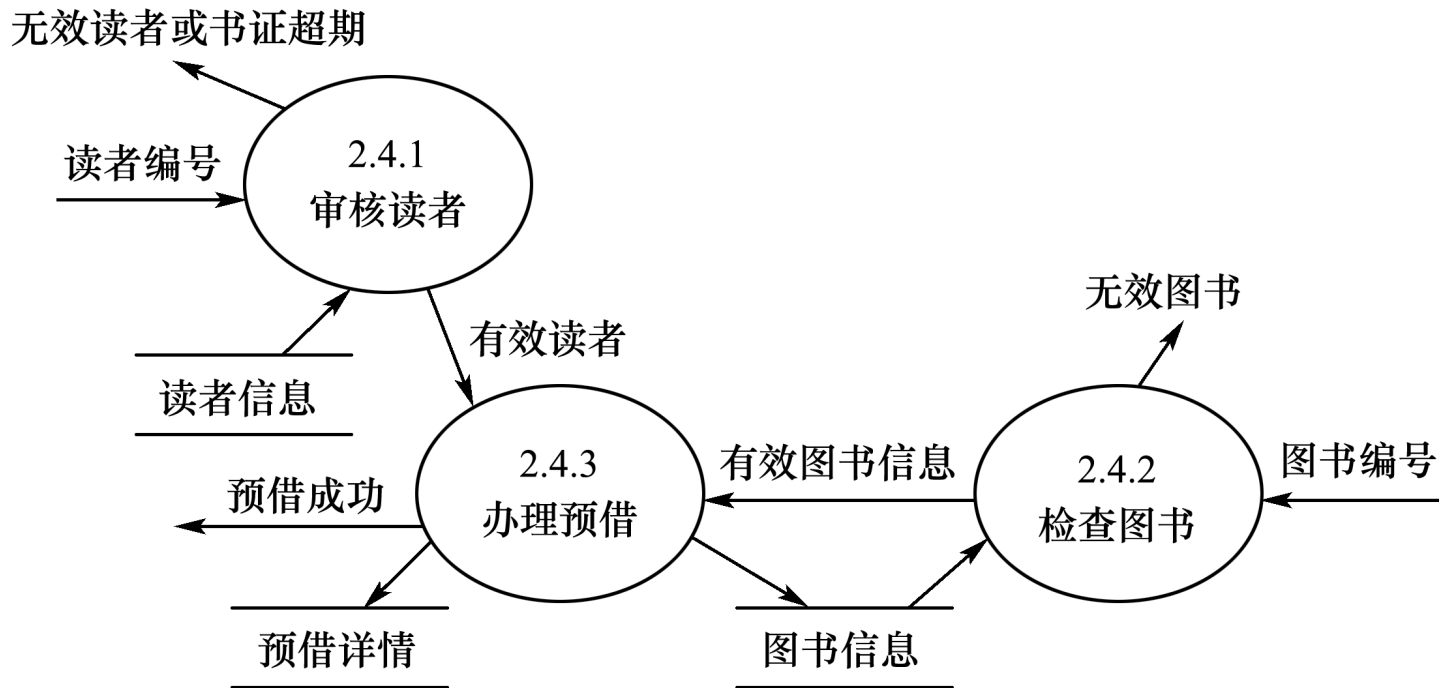
补充内容：结构化需求分析方法



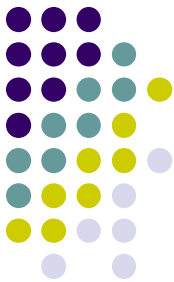
(f) 借阅管理加工的二层数据流图(还书)



补充内容：结构化需求分析方法



(g) 借阅管理加工的二层数据流图(预借)



补充内容：结构化需求分析方法

（2）数据词典

1. 数据词典的内容

一般说来，数据词典的每个条目中应包括以下信息。

- （1）名字：数据流、数据项、数据存储或外部实体的名称。
- （2）别名或编号：第（1）项中对象的其他名字。
- （3）分类：数据流、数据项、加工、数据存储、外部实体等。
- （4）内容描述：描述内容或数据结构等。
- （5）何处使用：哪些加工使用该条目。



补充内容：结构化需求分析方法

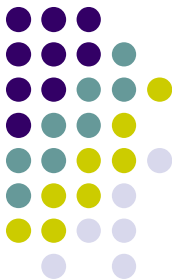
（2）数据词典

2. 数据词典中使用的符号

表 3-2

数据词典中常用符号

符号	含义	举例及说明
=	定义为	
+	与,连接	$x=a+b$ 表示 x 由 a 和 b 组成
$[\dots \dots]$	选择	$x=[a b c]$ 表示 x 由 a 或 b 或 c 组成,如:性别= $[男 女]$,表示性别是“男”或“女”之一
$\{ \dots \}$	重复	$x=\{a\}$ 表示 x 由 0 个或多个 a 组成,如:姓名= $1\{\text{汉字}\}4$,表示姓名为 1 至 4 个汉字; 又如: $1\{\text{字母}\}$,表示至少 1 个字母; $\{\text{字母}\}$,表示 0 至任意个字母
$(\dots)$	可选	$x=(a)$ 表示 a 可在 x 中出现,也可不出现
..	连接符	$x=1..9$ 表示 x 可取 1 到 9 任意一个值
* ... *	注释	表示两个 * 之间的内容是对条目的注释



补充内容：结构化需求分析方法

（2）数据词典

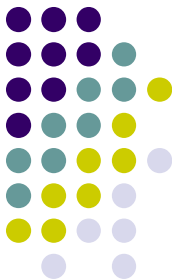
3. 数据词典书写实例

（1）“入库通知单”数据流的定义：

数据流名：入库通知单
数据流来源：“审查申请”加工
数据流去向：“图书入库”加工
组成：采购者＋采购者部门＋{图书详情＋数量}

（2）“入库通知单”数据流中的“图书详情”数据结构的定义：

名称：图书详情
组成：书名＋作者＋出版社＋单价＋购书日期



补充内容：结构化需求分析方法

（2）数据词典

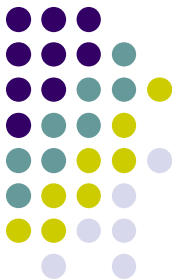
3. 数据词典书写实例

（3）“入库通知单”数据流中的“数量”数据项的定义：

数据项名：数量
别名：无
取值范围：正整数

（4）数据存储“图书库”的定义：

名称：图书库
组成：{图书编号+图书详情+目前库存量}
组织方式：按图书编号从小到大排列



补充内容：结构化需求分析方法

(2) 数据词典

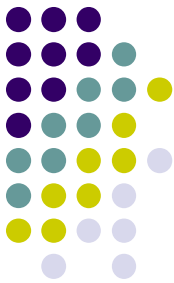
3. 数据词典书写实例

(5)外部实体“教师”的定义：

名称:教师
简述:向教师图书室提供图书的教师
从外部实体输入:报销申请
向外部实体输出:入库证明

(6)“审查申请”加工的定义：

名称:审查申请
输入:报销申请、专业教师登记表、专项资金
输出:拒绝信息或入库通知单
加工逻辑:IF 报销人员是专业教师且专项资金未用完 THEN 可以报销 ELSE 拒绝报销



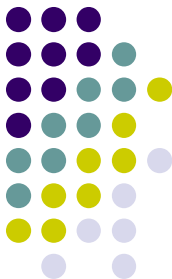
补充内容：结构化需求分析方法

（2）数据词典

4. 数据词典的实现

不论通过哪种途径实现的数据词典都应尽量做到以下几点：

- (1) 没有冗余：主要指数据定义不能重复。在规格说明书的其他组成部分中已出现的信息不能重复。
- (2) 查阅方便：通过名字可以方便地查阅数据词典中的每个定义。
- (3) 定义的书写方法简单、方便、严谨，而且可读性强。
- (4) 建议采用卡片形式书写。



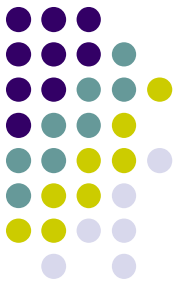
补充内容：结构化需求分析方法

(3) 加工逻辑的描述

1. 结构化语言

表 3-3 结构化语言

	书写格式	含义
选择结构	IF 〈条件〉 THEN 〈策略〉	如果 〈条件〉 则 〈策略〉
	IF 〈条件〉 THEN 〈策略 1〉 ELSE 〈策略 2〉 按下列条件选择策略	如果 〈条件〉 则 〈策略 1〉 否则 〈策略 2〉 按下列条件选择策略
	CASE 〈条件 1〉 〈策略 1〉 CASE 〈条件 2〉 〈策略 2〉 ... CASE 〈条件 N〉 〈策略 N〉	情况 〈条件 1〉 〈策略 1〉 情况 〈条件 2〉 〈策略 2〉 ... 情况 〈条件 N〉 〈策略 N〉
	WHILE 〈条件〉 DO 〈策略〉 END DO	每当 〈条件〉 成立则 〈策略〉 返回
	REPEAT 〈策略〉 UNTIL 〈条件〉	重复以下操作 〈策略〉 直到 〈条件〉



补充内容：结构化需求分析方法

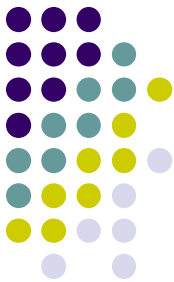
（3） 加工逻辑的描述

2. 判定表

在一些数据处理中，数据流图的加工需要经过多个逻辑条件组合的取值而确定，此时用自然语言或结构化语言难以描述，而运用判定表描述就比较清晰明了。

条件	条件取值的组合
动作	相应条件组合下应执行的动作

图 3-7 判定表的结构



补充内容：结构化需求分析方法

(3) 加工逻辑的描述

3. 判定树

判定树也是用来表达加工逻辑的工具，它是判定表的变形，有时比判定表更直观，更易于理解和使用。图书优惠政策的判定树如图 3-8 所示。

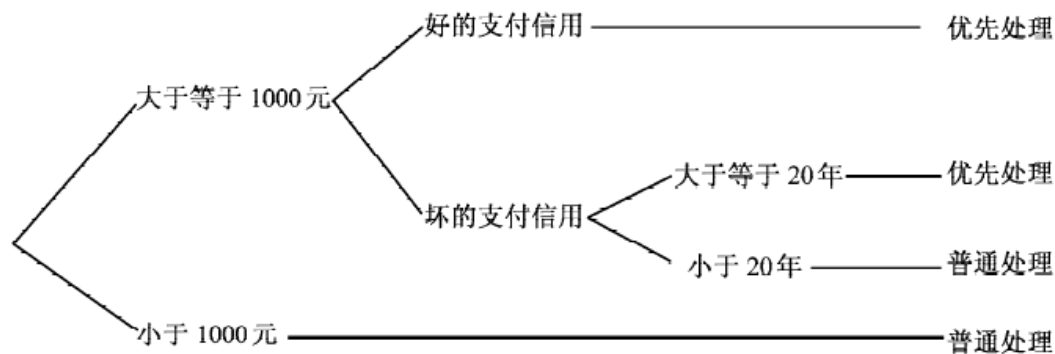
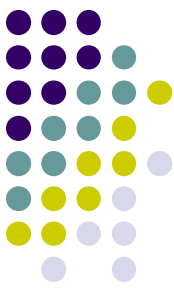


图 3-8 图书优惠政策判定树



补充内容：结构化需求分析方法

（4）需求规格说明书

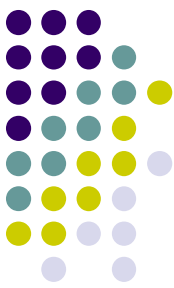
1. 需求规格说明书的编写内容

需求分析阶段应交付的主要文档是软件需求规格说明书。它提供了用户与开发人员对开发软件的共同理解，其作用相当于用户与开发单位之间的技术合同，是后续设计和编码的基础，是测试和验收的依据。

软件需求规格说明书的内容框架可参阅 GB/T 8567-2006 《计算机软件文档编制规范》。

在编写需求规格说明书时应注意以下几个问题：

- （1）说明书中的每一部分都非常重要，因此要慎重对待。
- （2）问题的描述要做到准确无误，没有二义性。
- （3）说明书的书写形式要规范。
- （4）允许用户根据项目的具体情况适当的将书写内容进行调整和筛选。



补充内容：结构化需求分析方法

(4) 需求规格说明书

2. 需求分析的评审

在需求分析规格说明书编写完成后，必须进行需求评审，以验证需求的正确性。

如果在评审过程中发现说明书存在错误或缺陷，应及时进行更改或弥补，重新进行相应部分的需求分析、需求建模、修改需求规格说明书，并再行评审。

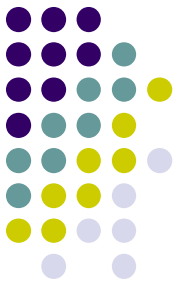
需求分析评审的主要内容如下：

(1) 一致性。所有需求必须是一致的，任何一条需求不能和其他需求相矛盾。

(2) 完整性。需求必须是完整的，规格说明书应该包括用户需要的每一个功能或性能。

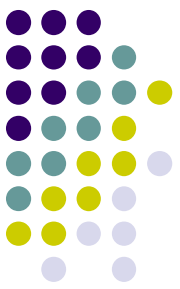
(3) 现实性。指定的需求应该是用现有的软硬件技术基本上可以实现的。对硬件技术的进步可以预测，对软件技术的进步则很难预测，只能从现有技术水平判断需求的现实性。

(4) 有效性。必须证明需求是正确而有效的，确实能解决用户所面对的问题。



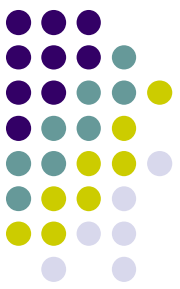
Summary

- Systems analysis involves defining system requirements— **functional** and **non-functional**
- Analysis activities include (分析活动包括)
 - Gather detailed information (收集信息)
 - Define requirements (定义需求)
 - Prioritize requirements (优先需求)
 - Develop user-interface dialogs (用户界面)
 - Evaluate requirements with users (评估需求)
- FURPS+ is the acronym for functional, usability, reliability, performance, and security requirements



Summary

- Models and modeling are used to explore and document requirements (模型和建模用于探索和记录需求)
- A model represents some aspect of a system, and can include textual, graphical, and mathematical models (模型包括文本、图形、数据模型)
- Unified Modeling Language (UML) is the standard set of notations and terminology for information systems models (统一建模语言 UML 是标准集合)



Summary

- Stakeholders are the people who have an interest in the success of the project (利益相关人是对项目成功感兴趣的人)
- There are **internal** vs. **external** stakeholders (内部 vs 外部相关者) and **operational** vs. **executive** stakeholders (运营 vs 执行相关者)
- Information gathering techniques are used to collect information about the project (信息收集)
 - Interviews, questionnaires, reviewing documents, observing business processes, researching vendors, comments and suggestions (访谈、问卷、文件、观察业务流程等)